CMDiff-Reg: Geometry-Enhanced Diffusion Intermediate Representations for Unsupervised Cross-Modality Remote Sensing Registration

Xiaochen Wei, Weiwei Guo, *Member, IEEE*, Wenxian Yu, *Senior Member, IEEE*, and Feiming Wei, *Member, IEEE*

***Abstract*—Cross-modality image registration in remote sensing remains a challenging problem due to the large radiometric discrepancies across heterogeneous sensors and the lack of precisely aligned training annotations. Existing unsupervised approaches typically rely on indirect supervision based on feature similarity or image translation. However, similarity-based methods often fail to accurately capture true spatial displacement under severe modality gaps, while translation-based strategies may suppress geometric structures or introduce artifacts due to the absence of reliable supervision. To address these challenges, we propose CMDiff-Reg, an unsupervised cross-modality remote sensing registration framework based on diffusion-denoised intermediate representations (DDIRs). Instead of explicitly translating images across modalities, CMDiff-Reg introduces diffusion-generated intermediate representations that serve as an auxiliary modality for self-supervised learning. Specifically, we design a geometry-enhanced conditional diffusion model that incorporates displacement-sensitive geometric conditions into the denoising process. The diffusion model is trained with precise noise supervision, enabling the generated representations to preserve some geometric structures while introducing significant appearance variations across modalities. By leveraging these geometry-consistent yet appearance-diverse representations, CMDiff-Reg constructs challenging self-supervised cross-modal training pairs with known geometric transformations, allowing the registration network to directly learn displacement estimation under large modality discrepancies. This design effectively avoids unstable pseudo-supervision in translation-based frameworks and improves training stability and representation robustness. Extensive experiments on multiple cross-modality benchmarks demonstrate that CMDiff-Reg consistently outperforms state-of-the-art unsupervised registration methods and achieves performance competitive with advanced supervised approaches, particularly under challenging modality discrepancies. These results highlight the potential of diffusion-based intermediate representations as a reliable self-supervision mechanism for cross-modality remote sensing image registration.**

***Index Terms*—Denoise signal, diffusion model, multimodal image registration, self-supervision, unsupervised learning.**

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I. Introduction

CROSS-MODALITY image registration plays a crucial role in remote sensing by aligning images captured from heterogeneous sensors through estimating spatial transformations that bridge modality discrepancies. Accurate cross-modal alignment establishes consistent geometric correspondence across sensing modalities and serves as a fundamental prerequisite for many remote sensing applications, including change detection [1], multimodal data fusion [2], cross-domain object detection [3], and geo-localization [4]. In these applications, even small alignment errors can significantly degrade downstream performance, highlighting the importance of robust cross-modality image registration.

With the rapid development of deep learning, end-to-end learning-based registration approaches [5], [6], [7], [8], [9], [10] have achieved substantial performance improvements by learning discriminative representations and directly estimating geometric transformations between image pairs. However, most existing methods rely on supervised training with precisely aligned image pairs or accurate geometric annotations. In practical remote sensing scenarios, such annotations are difficult to obtain because they require labor-intensive manual labeling or accurate estimation of transformation fields across heterogeneous modalities. As a result, the dependence on precise supervision limits the applicability of supervised registration methods in real-world cross-modality scenarios.

To reduce the reliance on annotated data, unsupervised learning has emerged as an important alternative. As illustrated in Fig. 1(a), many existing methods [11], [12], [13] employ feature projectors $\mathcal{P}$ to map cross-modality images into a modality-invariant feature space. The registration network $\mathcal{R}^C$ is then optimized by minimizing the discrepancy between the source image and the warped target image. Although this strategy alleviates the need for annotations, it relies on indirect supervision based on intensity or feature similarity [14]. In cross-modality remote sensing scenarios, where heterogeneous sensors exhibit significantly different imaging mechanisms and radiometric characteristics, such similarity-based losses are inherently ambiguous and often fail to capture true geometric correspondence, resulting in unstable optimization and inaccurate displacement estimation [14].

To provide more direct geometric supervision, recent studies [14], [15] have explored self-supervised registration strategies using synthetic image pairs with known geometric transfor-

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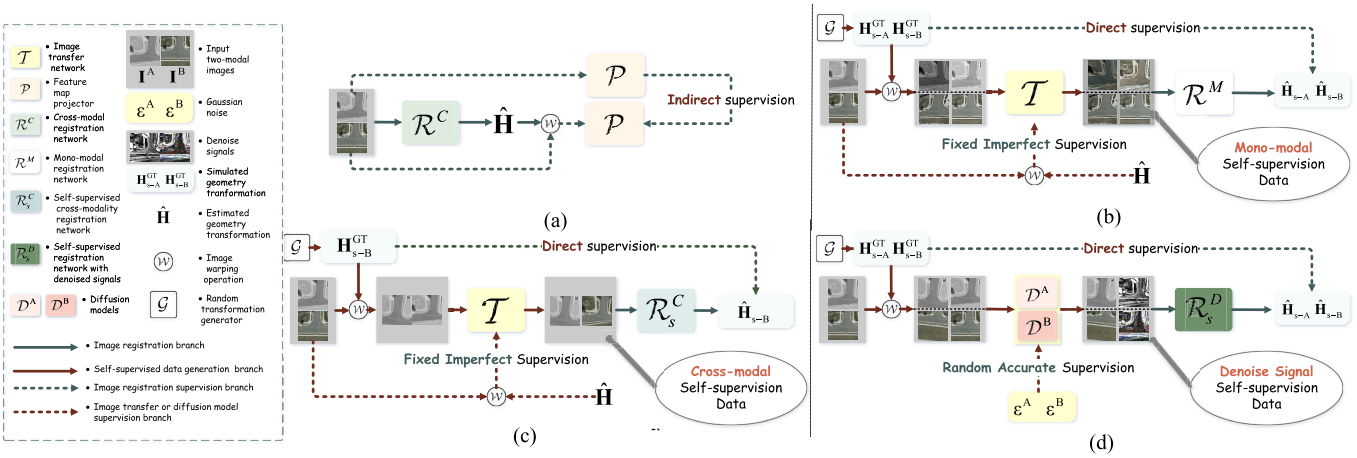


Fig. 1. Unsupervised multimodal image registration methods. (a) MU-Net, CHA-Net, AltO: registration with indirect supervision. (b) SSHNet: registration using monomodal self-supervised pairs. (c) CoLReg: registration with cross-modal self-supervised pairs. (d) CMDiff-Reg: registration via DDIRs for self-supervised data generation.

mations. As illustrated in Fig. 1(b), SSHNet [14] adopts a dual-network framework in which a monomodal registration network $\mathcal{R}^M$ is trained using synthetic image pairs, while a translation network $\mathcal{T}$ learns source-to-target mapping using pseudo ground truth generated by warping the target image with estimated transformations. Although this strategy introduces displacement-level supervision, its effectiveness is highly dependent on the quality of cross-modal translation, which may still retain modality discrepancies.

To further alleviate modality sensitivity, CoLReg [15], shown in Fig. 1(c), constructs cross-modal self-supervised training pairs using an image translation network and optimizes a cross-modality registration network $\mathcal{R}_s^C$. The quality of self-supervised samples still depends heavily on the translation network. In practice, these translation-based frameworks often lack explicit geometric constraints, making them prone to significantly suppress high-frequency structural details or introduce severe artifacts, especially under large modality discrepancies.

Despite these advances, two key challenges remain in unsupervised cross-modality registration. First, existing methods often highly rely on inaccurate transformation estimates, especially during early training stages, leading to unstable optimization and degraded geometric representation learning. Second, most approaches construct training pairs within fixed modality mappings, limiting the exploration of a broader modality space and making the learned representations sensitive to specific modality combinations.

Fundamental Technical Gap: Existing unsupervised cross-modality registration frameworks lack a stable self-supervision mechanism that is independent of modality-specific appearance. Current methods either rely on indirect similarity-based constraints, which are unreliable across heterogeneous modalities, or depend on translation-based self-supervision, which is sensitive to intermediate generation quality. This fundamental limitation prevents existing approaches from achieving robust and accurate geometric alignment under large modality discrepancies.

To address these limitations, we propose **CMDiff-Reg**, an unsupervised cross-modality remote sensing registration framework based on **diffusion-denoised intermediate representations (DDIRs)**. Unlike existing similarity- or translation-based frameworks, CMDiff-Reg introduces diffusion-generated intermediate representations as an auxiliary modality for self-supervised learning. Specifically, the stochastic denoising process of diffusion models under controlled noise injection is exploited to generate representations that preserve some geometric structures while introducing significant appearance variations. This enables the construction of geometry-preserved training pairs without relying on conventional image-to-image translation. These representations act as an auxiliary modality, enabling the construction of cross-modal training pairs with known geometric transformations.

To ensure geometric reliability, we further design a **geometry-enhanced conditional diffusion model** that incorporates displacement-sensitive structural conditions into the diffusion denoising process. The diffusion model is trained with precise noise supervision, providing explicit and stable constraints during training, which avoids the instability caused by pseudo-supervision in translation-based frameworks while preserving geometric structures essential for alignment. By combining geometry-enhanced diffusion representations with self-supervised transformation learning, CMDiff-Reg enables robust displacement estimation under large modality discrepancies.

The main contributions of this work are summarized as follows.

1) *DDIR Framework:* We propose CMDiff-Reg, an unsupervised framework that introduces DDIRs as an auxiliary modality. By constructing self-supervised training pairs without requiring ground-truth cross-modal alignment, CMDiff-Reg enables the registration network to learn robust geometric representations and reduces dependence on specific modality mappings.
2) *Geometry-Enhanced Conditional Diffusion Model:* We design a geometry-enhanced conditional diffusion model

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that integrates geometric condition extractors with structure-aware constraints and noise-domain supervision, preserving some displacement-sensitive geometric structures while avoiding instability and artifacts introduced by translation-based pseudo supervision.

3) *Diffusion-Driven Construction of Challenging Self-Supervised Cross-Modality Training Pairs:* By leveraging the stochastic denoising property of diffusion models under controlled noise injection, CMDiff-Reg generates intermediate representations that maintain structural consistency while introducing substantial appearance discrepancies, enabling robust displacement learning under large modality gaps.
4) *Extensive Experimental Validation on Multiple Cross-Modality Benchmarks:* Comprehensive experiments on several challenging datasets demonstrate that CMDiff-Reg outperforms state-of-the-art unsupervised registration methods and achieves performance competitive with some advanced supervised approaches.

II. Related Works

A. Supervised Deep Learning Methods

Current supervised deep learning methods can be divided into two paradigms: detector-based methods and detector-free methods. Therefore, we provide a concise review of these two paradigms.

Detector-based methods rely on detecting keypoints and learning descriptors or improving detection–description coupling for cross-modal matching. Representative approaches include local descriptor learning methods, such as HardNet [16], SosNet [17], HyNet [18], and SDGMNet [19], as well as multimodal matching methods, including MAP-Net [20], SDNet [21], and ReDFeat [22], which improve robustness to varying degrees in multimodal scenarios.

In contrast, detector-free methods bypass explicit keypoint detection and directly infer either pixel correspondences or transformation parameters using deep networks.

Correspondence-based methods [23], [24] establish dense or semi-dense correspondences, significantly improving robustness under large viewpoint changes, low-texture regions, and cross-modal conditions. Building upon this paradigm, recent works focus on improving feature interaction efficiency and matching reliability. EcoMatcher [25] replaces Transformer-based global interaction with a clustering-based representation (Context Clusters), enabling efficient long-range feature interaction while balancing computational cost and performance. In parallel, Xiang et al. [26] propose a detector-free framework that performs pixelwise dense correspondence search under a pseudo-epipolar constraint, followed by local refinement for subpixel accuracy. Zhang et al. [27] further integrate CNN-based multiscale features with self- and cross-attention and enforce consistency-based correspondence learning to improve robustness under severe radiometric and geometric variations. More recently, Li et al. [28] introduce a detector-free matching framework based on an interactive Swin Transformer, which leverages window-based cross-attention for efficient cross-modal feature interaction and adopts a coarse-to-fine hierarchical pipeline to generate accurate semidense correspondences.

In addition to correspondence-based approaches, end-to-end methods [8], [9], [10] formulate registration as a regression problem, where networks directly predict the geometric transformation between image pairs. Along this line, Hu et al. [29] approach registration as a decision-making process using CNN-based feature extraction and reinforcement learning for transformation estimation. Subsequent works further improve estimation accuracy through coarse-to-fine refinement strategies. AGNet [30] introduces a coarse-to-fine homography estimation framework that combines an adaptive query module with a gated interaction mechanism to progressively refine alignment accuracy, while CrossHomo [31] jointly models multimodal super-resolution and homography estimation within a multilevel coarse-to-fine framework to address both modality and resolution discrepancies.

B. Unsupervised Deep Learning Methods

Unlike supervised approaches that rely on precise ground-truth correspondences, unsupervised methods aim to align source and target images without explicit supervision by leveraging indirect cues or specialized loss functions. Some works [11], [12], [13], [32] construct a modality-invariant feature space to capture high-level discrepancies between warped image pairs. Another line of research [33], [34], [35], [36] employs image translation techniques to map source images into the target modality, enabling the use of monomodal similarity losses for supervision.

Recently, diffusion models [37], [38], [39], [40] have emerged as a powerful generative paradigm and have been increasingly explored for image registration. DiffuseMorph [41] utilizes a conditional diffusion model to learn deformation-aware latent representations and incorporates a deformation network to estimate dense registration fields. More recent works shift the focus toward exploiting diffusion models for data generation to alleviate the lack of reliable supervision. DMHomo [42] leverages diffusion models to synthesize realistic image pairs with accurate geometric annotations by learning from unlabeled data with pseudo-labels. This enables the construction of large-scale, high-quality training datasets for homography estimation, effectively mitigating the scarcity of labeled data. Similarly, CoLReg [15] incorporates a diffusion-based image translation module to generate modality-consistent image pairs, which serve as a reliable source of self-supervision. By progressively improving the quality of generated data and pseudo-labels through a collaborative training scheme, it enhances the robustness and accuracy of multimodal image registration under significant appearance variations.

Unsupervised methods [14], [15], [43] with self-supervision of displacement further improve performance by generating synthetic training pairs with known geometric transformations and directly optimizing networks using displacement-based loss functions, thereby enhancing both robustness and accuracy. SCPNet [43] constructs intramodality synthetic pairs via simulated homographies and enforces consistency through feature correlation and projection constraints. SSHNet [14]

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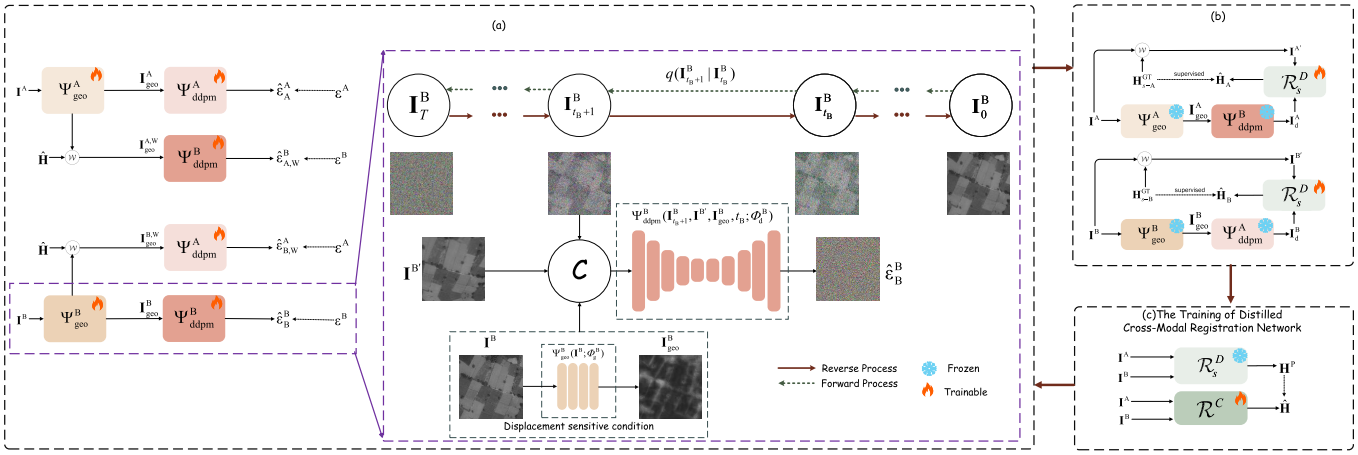


Fig. 2. Overview of the proposed CMDiff-Reg framework for unsupervised cross-modality image registration. CMDiff-Reg introduces DDIRs as the auxiliary modality to construct self-supervised training pairs with known geometric transformations. (a) Training of geometry-enhanced diffusion models $\mathcal{D}_{\mathrm{A}}$ and $\mathcal{D}_{\mathrm{B}}$ to generate structure-preserving intermediate representations. (b) Training of the denoised-representation self-supervised registration network $\mathcal{R}_s^D$ using geometrically supervised synthetic pairs. (c) Distillation to train the cross-modal registration network $\mathcal{R}^C$ for real unaligned image pairs.

reformulates unsupervised cross-modal homography estimation into two directly supervised subtasks and adopts a split optimization strategy with feature-space supervision to improve training stability and accuracy. CoLReg [15] combines synthetic supervision and real cross-modal data in a collaborative training manner, further improving generalization performance. Despite these advances, existing unsupervised cross-modal registration methods still suffer from a fundamental limitation: the generation of self-supervised samples typically relies on cross-modal consistency mappings or image-to-image translation networks, which lack direct supervision from precise ground-truth labels. This inevitably constrains the quality of synthesized training pairs and leads to suboptimal training stability. To address this issue, we propose CMDiff-Reg, which leverages denoising representations from a diffusion model trained with precise (or controllable-noise) supervision to construct high-quality cross-modal self-supervised samples. By introducing diffusion-based supervision into the sample generation process, our method significantly improves both training stability and registration performance.

C. Multimodal and Cross-Modal Learning in Remote Sensing

Recent advances in multimodal and cross-modal learning have significantly improved remote sensing tasks such as classification and semantic segmentation. For example, CrossCity Matters introduces a multimodal benchmark dataset and domain adaptation framework for cross-city semantic segmentation [44], while CNN-based methods have been widely explored for multimodal remote sensing data classification through feature fusion and representation learning [45]. Other studies further investigate multimodal fusion and cross-modal representation learning [46], [47], [48]. However, these approaches mainly focus on semantic-level understanding and typically assume spatially aligned data, without explicitly addressing geometric discrepancies across modalities. In contrast, our work focuses on cross-modality image registration, where accurate geometric alignment is the primary objective.

III. Methodology

A. Overview

Given an unaligned cross-modality image pair $\{\mathbf{I}^{\mathrm{A}}, \mathbf{I}^{\mathrm{B}}\}$, the goal of cross-modality image registration is to estimate the geometric transformation that aligns the source image with the target image. However, in most practical remote sensing scenarios, the ground-truth transformation between heterogeneous modalities is unavailable, making direct supervised training difficult. To address this challenge, we propose CMDiff-Reg, an unsupervised cross-modality registration framework that introduces DDIR as an auxiliary modality for self-supervised training. As illustrated in Fig. 2, the proposed framework consists of three training stages.

1) Training Geometric-Enhanced Diffusion Models: First, we train two geometry-enhanced diffusion models $\mathcal{D}_{\mathrm{A}}$ and $\mathcal{D}_{\mathrm{B}}$ for different modalities. Each diffusion model consists of a geometric condition extractor $\Psi_{\mathrm{geo}}^{\mathrm{M}}(\mathrm{M} \in \{\mathrm{A}, \mathrm{B}\})$ and a conditional DDPM network $\Psi_{\mathrm{ddpm}}^{\mathrm{M}}$. The geometric condition extractor learns displacement-sensitive structural representations, which guide the diffusion denoising process to generate intermediate representations that preserve some geometric structures while allowing appearance variations.

2) Training the Denoised-Representation Self-Supervised Registration Network: Using the trained diffusion models, denoised intermediate representations are generated from each modality. By applying known geometric transformations to the original images, cross-modal training pairs with precise geometric supervision are constructed. These pairs are then used to train a denoised-representation-based registration network $\mathcal{R}_s^D$ to directly learn displacement estimation.

3) Distillation Training of the Cross-Modal Registration Network: After the self-supervised registration network is trained, it is utilized to estimate pseudo-transformation labels

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for real unaligned cross-modality image pairs, following the general pseudo-labeling paradigm adopted in [14], [15]. These pseudo labels supervise the training of the final cross-modal registration network $\mathcal{R}^C$, which is used for real-world image alignment.

Through this three-stage training process, CMDiff-Reg enables stable self-supervised learning of geometric transformations while effectively handling large appearance discrepancies across sensing modalities.

B. Geometric-Enhanced Diffusion Models

Existing self-supervised cross-modal registration methods commonly rely on an image-to-image translation network $\mathcal{T}$ to synthesize target-domain images for constructing self-supervised training pairs. Although effective in reducing modality gaps, this strategy has two inherent limitations. First, the optimization of $\mathcal{T}$ depends on imperfect transformations estimated by the registration network. These estimates are often inaccurate at early training stages, which may cause unstable convergence. Second, the synthesized image pairs are restricted to fixed modality mappings, such as source-to-target translation. As a result, the registration network becomes sensitive to specific modality combinations and generalizes poorly.

To overcome these limitations, we introduce a geometry-enhanced conditional diffusion model that generates denoised representations as an auxiliary modality. At first glance, incorporating diffusion models may appear risky for geometric tasks, since diffusion processes are inherently stochastic and may blur structural details. This concern is valid for unconstrained image generation. In our framework, however, diffusion is not used for free image synthesis. Instead, it is tightly constrained by explicit geometric conditions and supervised in the noise domain using precise labels. This design confines stochasticity to appearance variations while preserving some displacement-sensitive geometric structures.

As illustrated in Fig. 3, for each modality $\mathrm{M} \in \{\mathrm{A}, \mathrm{B}\}$, we construct a geometric-enhanced diffusion model $\mathcal{D}_{\mathrm{M}}$. Each model consists of a geometric condition extractor $\Psi^{\mathrm{M}}_{\mathrm{geo}}$ and a conditional DDPM network $\Psi^{\mathrm{M}}_{\mathrm{ddpm}}$ based on a UNet architecture. The conditional DDPM network adopts a UNet architecture as the noise prediction backbone, following common practices in diffusion models [37], [39], [40]. Specifically, the network is implemented as a multiscale encoder–decoder with residual blocks and skip connections, where feature resolutions are progressively downsampled and upsampled to capture both global context and fine-grained details. The geometric condition map $\mathbf{I}^{\mathrm{M}}_{\mathrm{geo}}$ is injected into the diffusion model as an additional conditional input, which guides the noise prediction at each timestep. This design enables the diffusion process to preserve displacement-sensitive structures while allowing appearance variations. It is worth noting that the UNet backbone itself is not a contribution of this work. Instead, our focus lies in the proposed geometry-enhanced diffusion mechanism and the construction of DDIRs for self-supervised learning.

The extractor $\Psi^{\mathrm{M}}_{\mathrm{geo}}$ maps the input image to a geometric condition map $\mathbf{I}^{\mathrm{M}}_{\mathrm{geo}}$, which encodes displacement-sensitive structural information. The DDPM network $\Psi^{\mathrm{M}}_{\mathrm{ddpm}}$ predicts diffusion noise under the guidance of this condition. Importantly, the two diffusion models $\mathcal{D}_{\mathrm{A}}$ and $\mathcal{D}_{\mathrm{B}}$ share the same architecture but do not share parameters.

1) Training the Geometric Condition Extractor: For the itth co-training cycle, we first optimize the geometric condition extractor $\Psi^{\mathrm{M}}_{\mathrm{geo}}$. Our objective is twofold. First, $\Psi^{\mathrm{M}}_{\mathrm{geo}}$ should encode displacement-sensitive geometric cues (e.g., edges, contours, and junctions). Second, it should suppress modality-specific appearance patterns that are irrelevant to registration. This design is critical for diffusion-based self-supervision: diffusion denoising is stochastic in nature, and geometry can be compromised if the denoising process is unconstrained. In our framework, geometry is preserved because the stochastic denoising is explicitly guided by $\mathbf{I}^{\mathrm{M}}_{\mathrm{geo}}$ and is supervised in the noise domain with exact labels. To learn such geometry-aware conditions, we impose three complementary constraints, as shown in Fig. 3(a).

a) Cross-modal geometric consistency: We first enforce that the geometric condition extracted from modality M can effectively guide noise prediction in the other modality $\dot{\mathrm{M}}$. Intuitively, if $\mathbf{I}^{\mathrm{M}}_{\mathrm{geo}}$ captures modality-invariant structure, then it should remain useful even when applied to denoising the other modality. This constraint is formulated as

$$\begin{aligned}
\mathcal{L}_d^{g1\text{-}\mathrm{M}} &= \sum \left| \hat{\varepsilon}^{\dot{\mathrm{M}}}_{\mathrm{M,W}} - \varepsilon^{\dot{\mathrm{M}}} \right| \circ \mathrm{P}^{\mathrm{M}} \\
\mathbf{I}^{\mathrm{M,W}}_{\mathrm{geo}} &= \mathcal{W}\left(\mathbf{I}^{\mathrm{M}}_{\mathrm{geo}}, \hat{\mathbf{H}}^{\mathrm{M}\dot{\mathrm{M}}}\right) \\
\mathbf{I}^{\mathrm{M}}_{\mathrm{geo}} &= \Psi^{\mathrm{M}}_{\mathrm{geo}}\left(\mathbf{I}^{\mathrm{M}};\, \Phi_{\mathrm{g}}^{\mathrm{M}(it)}\right) \\
\mathbf{I}^{\dot{\mathrm{M}}}_{t_{\dot{\mathrm{M}}+1}} &= \mathcal{F}\left(\mathbf{I}^{\dot{\mathrm{M}}}, t_{\dot{\mathrm{M}}}+1, \bar{\alpha}_{t_{\dot{\mathrm{M}}}+1}, \varepsilon^{\dot{\mathrm{M}}}\right) \\
\hat{\varepsilon}^{\dot{\mathrm{M}}}_{\mathrm{MW}} &= \Psi^{\dot{\mathrm{M}}*}_{\mathrm{ddpm}}\left(\mathbf{I}^{\dot{\mathrm{M}}}_{t_{\dot{\mathrm{M}}}+1}, \mathbf{I}^{\dot{\mathrm{M}}\prime}, \mathbf{I}^{\mathrm{M,W}}_{\mathrm{geo}}, t_{\dot{\mathrm{M}}};\, \Phi_{\mathrm{d}}^{\dot{\mathrm{M}}*(it)}\right)
\end{aligned} \tag{1}$$

where $\Psi^{\dot{\mathrm{M}}*}_{\mathrm{ddpm}}$ is an auxiliary DDPM initialized from the previous co-training cycle. P^{M} is a binary mask that excludes padded regions caused by warping. $\mathcal{F}$ denotes the forward diffusion process that injects exact Gaussian noise ε^{M} at step $t_{\dot{\mathrm{M}}} + 1$. $\mathcal{W}(\cdot)$ denotes the geometric warping using the estimated transformation $\hat{\mathbf{H}}^{\mathrm{M}\dot{\mathrm{M}}}$. By enforcing accurate noise prediction under cross-modal geometric guidance, $\mathcal{L}_d^{g1\text{-}\mathrm{M}}$ encourages $\Psi^{\mathrm{M}}_{\mathrm{geo}}$ to learn geometry that generalizes across modalities.

b) Effectiveness of geometric guidance: The warped condition $\mathbf{I}^{\mathrm{M,W}}_{\mathrm{geo}}$ may be imperfect because $\hat{\mathbf{H}}^{\mathrm{M}\dot{\mathrm{M}}}$ is only an estimate. Such misalignment can lead to misleading conditions and unstable training if not controlled. To avoid harmful guidance, we enforce that conditioning should not degrade the noise prediction compared with an unguided baseline

$$\begin{aligned}
\mathcal{L}_d^{g2\text{-}\mathrm{M}} &= \sum \left(\left| \hat{\varepsilon}^{\dot{\mathrm{M}}}_{\mathrm{M,W}} - \varepsilon^{\dot{\mathrm{M}}} \right| - \left| \hat{\varepsilon}^{\dot{\mathrm{M}}}_{\mathrm{M}O} - \varepsilon^{\dot{\mathrm{M}}} \right| \right) \circ \mathrm{P}^{\mathrm{M}} \\
\hat{\varepsilon}^{\dot{\mathrm{M}}}_{\mathrm{M}O} &= \Psi^{\dot{\mathrm{M}}}_{\mathrm{ddpm}}\left(\mathbf{I}^{\dot{\mathrm{M}}}_{t_{\dot{\mathrm{M}}}+1}, \mathbf{I}^{\dot{\mathrm{M}}}, \mathbf{O}, t_{\dot{\mathrm{M}}};\, \Phi_{\mathrm{d}}^{\dot{\mathrm{M}}(it-1)}\right)
\end{aligned} \tag{2}$$

where $\mathbf{O}$ denotes an all-constant condition map. This term explicitly regularizes the condition extractor by preventing it from producing geometric guidance that is detrimental to denoising.

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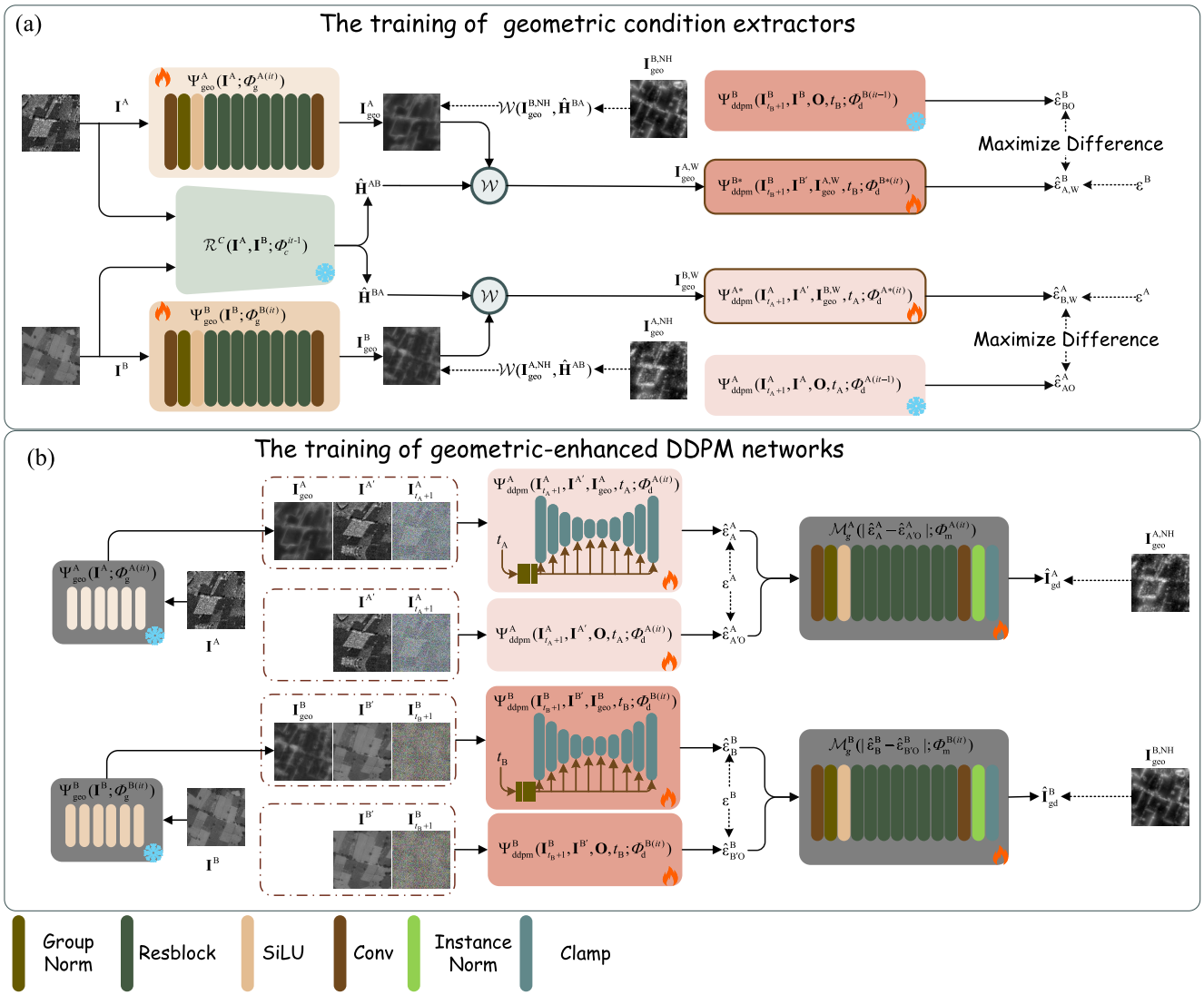


Fig. 3. Training process of the proposed geometry-enhanced conditional diffusion models in CMDiff-Reg. For each modality $\mathrm{M} \in \{\mathrm{A}, \mathrm{B}\}$, the diffusion model consists of a geometric condition extractor $\Psi^{\mathrm{M}}_{\mathrm{geo}}$ and a conditional DDPM network $\Psi^{\mathrm{M}}_{\mathrm{ddpm}}$. (a) Optimization of the geometric condition extractors, which learn displacement-sensitive structural representations to guide the diffusion process. (b) Optimization of the conditional DDPM networks using geometry-aware conditions to generate diffusion-denoised representations while preserving some geometric structures.

c) Explicit geometric structure alignment: Beyond cross-modal usefulness, we further enforce that the extracted geometric condition preserves structural fidelity. Specifically, we align $\mathbf{I}^{\mathrm{M}}_{\mathrm{geo}}$ with the warped geometric representation of the other modality. This directly encourages structural consistency across modalities

$$\mathcal{L}_d^{g3\text{-M}} = \sum \left| \mathbf{I}^{\mathrm{M}}_{\mathrm{geo}} - \mathbf{I}^{\dot{\mathrm{M}},\mathrm{NH,W}}_{\mathrm{geo}} \right| \circ \mathrm{P}^{\dot{\mathrm{M}}}$$
$$\mathbf{I}^{\dot{\mathrm{M}},\mathrm{NH,W}}_{\mathrm{geo}} = \mathcal{W}\left(\mathbf{I}^{\dot{\mathrm{M}},\mathrm{NH}}_{\mathrm{geo}}, \hat{\mathbf{H}}^{\dot{\mathrm{M}}\mathrm{M}}\right) \tag{3}$$

where $\mathbf{I}^{\dot{\mathrm{M}},\mathrm{NH}}_{\mathrm{geo}}$ is a normalized handcrafted geometric feature inspired by [49]. This constraint provides a stable geometric anchor, especially at the beginning of co-training, and encourages $\Psi^{\mathrm{M}}_{\mathrm{geo}}$ to focus on geometry rather than modality-specific textures.

d) Overall objective: The overall objective for optimizing the geometric condition extractor is

$$\mathcal{L}_d^{g\text{-M}} = \lambda_{g1}\mathcal{L}_d^{g1\text{-M}} + \lambda_{g2}\mathcal{L}_d^{g2\text{-M}} + \lambda_{g3}\mathcal{L}_d^{g3\text{-M}}. \tag{4}$$

2) Training the Geometric-Enhanced DDPM: After learning the geometric conditions, we train the conditional DDPM network $\Psi^{\mathrm{M}}_{\mathrm{ddpm}}$ under aligned geometric guidance, as shown in Fig. 3(b). This stage is responsible for producing denoised representations that serve as the pivot modality. A key concern is whether diffusion denoising may smooth out geometric structures. In our design, geometric structure is preserved for two reasons. First, the model is supervised with exact noise labels, which avoids drifting solutions caused by noisy pseudo targets. Second, the denoising behavior is constrained by $\mathbf{I}^{\mathrm{M}}_{\mathrm{geo}}$ and further regularized by an explicit geometric recovery loss.

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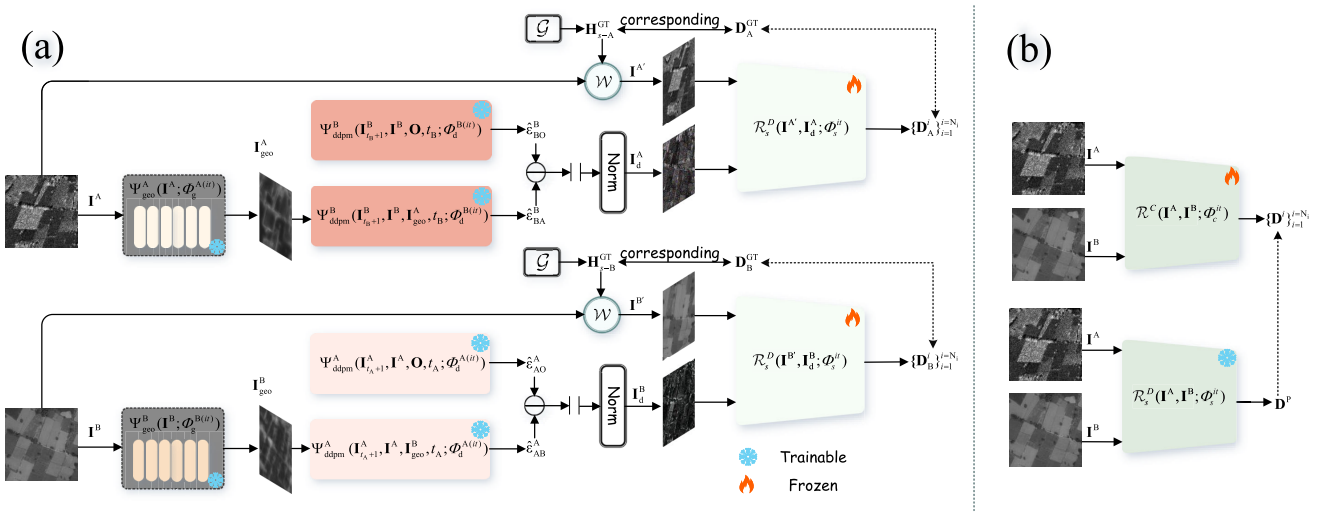


Fig. 4. (a) Training of self-supervised registration network $\mathcal{R}_s^D$ with denoised intermediate representations training samples $\{\mathbf{I}^{A'}, \mathbf{I}_d^A, \mathbf{D}_A^{GT}\}$ and $\{\mathbf{I}^{B'}, \mathbf{I}_d^B, \mathbf{D}_B^{GT}\}$. $\mathbf{D}_A^{GT}$ and $\mathbf{D}_B^{GT}$ are the displacements of four corners by parameterizing the matrix $\mathbf{H}_{s-A}^{GT}$ and $\mathbf{H}_{s-B}^{GT}$, respectively. $\{\mathbf{D}_A^i\}_{i=1}^{i=N_i}$ and $\{\mathbf{D}_B^i\}_{i=1}^{i=N_i}$ are the predicted displacements of corners, and N_i is the number of iterations for registration network. (b) Training process of the cross-modality registration network $\mathcal{R}^C(\Phi_c^{it})$ using unsupervised image pairs $\{\mathbf{I}^A, \mathbf{I}^B\}$ at cycle *it*. The pseudo-label $\mathbf{D}^P$ obtained from the intermediate network is used to supervise the predicted corner displacements $\{\mathbf{D}^i\}_{i=1}^{i=N_i}$.

To emphasize structure-sensitive regions, we weight the diffusion loss with a geometric importance map $\mathbf{I}_{geo}^{M,H}$

$$\mathcal{L}_d^{dw\text{-}M} = \sum \left(\mathbf{I}_{geo}^{M,H} \odot \|\hat{\boldsymbol{\varepsilon}}_M - \boldsymbol{\varepsilon}_M\|_1\right)$$
$$\hat{\boldsymbol{\varepsilon}}_M^M = \Psi_{ddpm}^M\left(\mathbf{I}_{t_M+1}^M,\ \mathbf{I}^{M'},\ \mathbf{I}_{geo}^M,\ t_M;\ \Phi_d^{M(it)}\right) \tag{5}$$

where $\mathbf{I}_{geo}^{M,H}$ is handcrafted geometric feature inspired by [49]. This weighting encourages the model to fit the noise more accurately around salient geometric structures, rather than focusing on modality-dependent texture patterns.

To explicitly preserve the geometric structure in the denoised representations, we further enforce geometric recovery consistency with the learnable geometric metric $\mathcal{M}_g^M$

$$\mathcal{L}_d^{dg\text{-}M} = \sum \left|\hat{\mathbf{I}}_{gd}^M - \mathbf{I}_{geo}^{M,NH}\right|$$
$$\hat{\mathbf{I}}_{gd}^M = \mathcal{M}_g^M\left(\left|\hat{\varepsilon}_M^M - \hat{\varepsilon}_{M'O}^M\right|;\ \Phi_m^{M(it)}\right) \tag{6}$$

where the term $|\hat{\varepsilon}_M^M - \hat{\varepsilon}_{M'O}^M|$ captures the structural effect introduced by the geometric condition. By mapping this difference back through $\mathcal{M}_g^M$, we explicitly constrain the diffusion model to preserve the geometric information encoded in $\mathbf{I}_{geo}^{M,NH}$. This mechanism is the one characteristic of our *geometric enhancement*: diffusion-induced stochasticity is confined to appearance variation, while geometry is preserved and reinforced through condition-guided denoising and geometric recovery supervision. The unguided noise prediction is learned via

$$\mathcal{L}_d^{do\text{-}M} = \sum \left|\hat{\varepsilon}_{M'O}^M - \varepsilon^M\right|. \tag{7}$$

Finally, the overall loss for optimizing the diffusion model is

$$\mathcal{L}_d^{d\text{-}M} = \lambda_{dw}\mathcal{L}_d^{dw\text{-}M} + \lambda_{dg}\mathcal{L}_d^{dg\text{-}M} + \lambda_{do}\mathcal{L}_d^{do\text{-}M}. \tag{8}$$

C. $\mathcal{R}_s^D$: Self-Supervised Registration With Denoised Representations

In this section, we will describe the training of the denoised-representations-based self-supervised registration network $\mathcal{R}_s^D$. The key idea is to construct cross-modal training pairs that exhibit large modality discrepancies while preserving geometric consistency.

1) Generation of Geometrically Supervised Image Pairs: As illustrated in Fig. 4(a), we first apply random geometric transformations $\mathbf{H}_{s-A}^{GT}$ and $\mathbf{H}_{s-B}^{GT}$ to the source and target images, respectively. This produces monomodal image pairs $\{\mathbf{I}^{A'}, \mathbf{I}^A\}$ and $\{\mathbf{I}^{B'}, \mathbf{I}^B\}$, for which the ground-truth geometric transformations are known.

To transform these monomodal pairs into cross-modal self-supervised pairs, we introduce denoised representations as an intermediate modality. Specifically, the geometric condition extracted from one modality is injected into the diffusion model of the other modality.

2) Denoised Representations Construction: For modality M, the denoised representation is computed as

$$\mathbf{I}_d^M = \text{Norm}\left\{\left|\hat{\varepsilon}_{\dot{M}O}^{\dot{M}} - \hat{\varepsilon}_{\dot{M}M}^{\dot{M}}\right|\right\}$$
$$\mathbf{I}_{geo}^M = \Psi_{geo}^M\left(\mathbf{I}^M; \Phi_g^{M(it)}\right)$$
$$\hat{\varepsilon}_{\dot{M}O}^{\dot{M}} = \Psi_{ddpm}^{\dot{M}}\left(\mathbf{I}_{t_{\dot{M}}+1},\ \mathbf{I}^{\dot{M}},\ \mathbf{O},\ t_{\dot{M}};\ \Phi_d^{\dot{M}(it)}\right)$$
$$\hat{\varepsilon}_{\dot{M}M}^{\dot{M}} = \Psi_{ddpm}^{\dot{M}}\left(\mathbf{I}_{t_{\dot{M}}+1},\ \mathbf{I}^{\dot{M}},\ \mathbf{I}_{geo}^M,\ t_{\dot{M}};\ \Phi_d^{\dot{M}(it)}\right) \tag{9}$$

where Norm is the instance normalization without learnable affine transformation. By taking the difference between guided and unguided noise predictions, the resulting denoised representations emphasizes the geometric contribution introduced by the condition. This operation preserves some geometric structures while amplifying appearance variations.

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3) Self-Supervised Training Objective: In this article, as shown in Fig. 5, we adopt MCNet [10], modified from its original three-scale configuration to incorporate four scales as the registration network The denoised representations $\mathbf{I}_{\mathrm{d}}^{\mathrm{A}}$ and $\mathbf{I}_{\mathrm{d}}^{\mathrm{B}}$ serve as the auxiliary modality. They are paired with geometrically transformed real images to form cross-modal self-supervised training pairs $\{\mathbf{I}^{\mathrm{A}'}, \mathbf{I}_{\mathrm{d}}^{\mathrm{A}}\}$ and $\{\mathbf{I}^{\mathrm{B}'}, \mathbf{I}_{\mathrm{d}}^{\mathrm{B}}\}$.

The registration network $\mathcal{R}_s^D$ is trained using direct displacement supervision

$$\mathcal{L}_s = \sum_{\mathrm{M}\in\{\mathrm{A},\mathrm{B}\}} \mathcal{L}_{\mathrm{disp}}\left(\{\mathbf{D}_{\mathrm{M}}^i\}_{i=1}^{i=\mathrm{N_i}}, \mathbf{D}_{\mathrm{M}}^{\mathrm{GT}}\right) \tag{10}$$

where $\{\mathbf{D}_{\mathrm{M}}^i\}_{i=1}^{i=\mathrm{N_i}}$ are the iterative displacement predictions produced by $\mathcal{R}_s^D$. The displacement loss $\mathcal{L}_{\mathrm{disp}}$ follows [10] and directly enforces geometric consistency, independent of modality appearance.

D. $\mathcal{R}^C$: Knowledge-Distilled Cross-Modal Image Registration

To adapt the registration model to real-world cross-modal data, we introduce a knowledge-distilled network $\mathcal{R}^C$. This network is trained using pseudo-supervision provided by the intermediate self-supervised model $\mathcal{R}_s^D(\Phi_s^{it})$, as illustrated in Fig. 4(b). Unlike $\mathcal{R}_s^D$, which is optimized on synthetic self-supervised pairs, $\mathcal{R}^C$ is explicitly trained on real, unlabeled cross-modal image pairs. Given a real image pair $\{\mathbf{I}^{\mathrm{A}}, \mathbf{I}^{\mathrm{B}}\}$, the self-supervised network $\mathcal{R}_s^D$ produces a sequence of displacement predictions. The final-stage output is selected as the pseudo displacement label $\mathbf{D}^{\mathrm{P}}$, which serves as surrogate supervision for training $\mathcal{R}^C$. In this way, the geometric knowledge learned from denoised self-supervision is transferred to a network specialized for real cross-modal registration. The objective function for optimizing $\mathcal{R}^C$ is defined as

$$\begin{aligned}
\mathcal{L}_c &= \mathcal{L}_{\mathrm{disp}}\left(\{\mathbf{D}^i\}_{i=1}^{i=\mathrm{N_i}}, \mathbf{D}^{\mathrm{P}}\right) \\
\{\mathbf{D}^i\}_{i=1}^{i=\mathrm{N_i}} &= \mathcal{R}^C\left(\mathbf{I}^{\mathrm{A}}, \mathbf{I}^{\mathrm{B}}; \Phi_c^{it}\right) \\
\mathbf{D}^{\mathrm{P}} &= \mathbf{D}_{\mathrm{N_i}}^{\mathrm{AB}} \\
\{\mathbf{D}_i^{\mathrm{AB}}\}_{i=1}^{i=\mathrm{N_i}} &= \mathcal{R}_s^D\left(\mathbf{I}^{\mathrm{A}}, \mathbf{I}^{\mathrm{B}}; \Phi_s^{it}\right).
\end{aligned} \tag{11}$$

E. Overflow of Training and Testing

Algorithm 1 introduces the details of training CMDiff-Reg. After the training, we obtained a cross-modality registration network $\mathcal{R}^C(\Phi_c^{(\mathrm{IT}-1)})$. For testing, we can utilize $\mathcal{R}^C$ to estimate the transformation matrix $\hat{\mathbf{H}}$.

IV. Experiment and Results

A. Experimental Setting

1) Datasets: We evaluate CMDiff-Reg on five cross-modality datasets covering remote sensing and general cross-modal scenarios.

a) GoogleEarth: Contains 8750 training and 850 testing unaligned image pairs from Google Earth imagery [50].

b) RGB_IR_AI: Derived from [51], this dataset includes 1210 aligned infrared–RGB pairs (512×512). After random perturbations [52], 9801 training and 1089 testing unaligned pairs are generated.

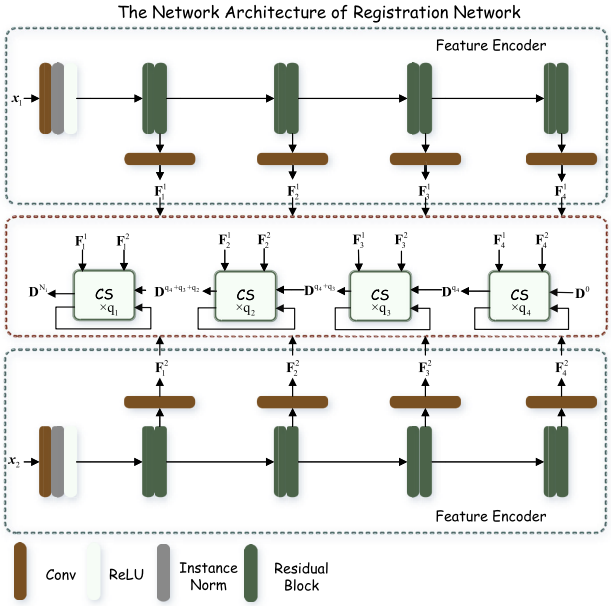


Fig. 5. Network architecture of the self-supervised and distilled registration networks. It adopts the same four-scale modified MCNet architecture [10], as used in [15].

c) Depth_VIS: Built from the DIML benchmark [53], it contains RGB–depth pairs from 18 indoor scenes. Using $10\times$ oversampling with random perturbations, we obtain 16 210 training and 4910 testing samples.

d) SAR_Opt_OS: Contains SAR–optical pairs from satellite observations [54]. After resizing to 192×192 and $5\times$ oversampling, 10 055 training and 2120 testing samples are obtained.

e) SAR_Opt_OS2: Similar to SAR_Opt_OS [55], this dataset uses nine locations for training and four for testing, resulting in 13 040 training and 5975 testing samples.

2) Evaluation Metrics: Three complementary metrics are used as follows.

1) *Average Corner Error (ACE):* Mean corner error between predicted and ground-truth displacements.
2) *AUC@k:* Proportion of samples with ACE below thresholds $k \in \{3, 5, 10, 20\}$.
3) *MACE:* Dataset-level average ACE.

3) Compared Methods: We compare our approach with representative state-of-the-art image registration methods, including as follows.

1) *Supervised*: DHN [52], IHN1 (CVPR 2022) [8] and IHN (CVPR 2022) [8], RHWF1 (CVPR 2023) [9] and RHWF (CVPR 2023) [9], and MCNet (CVPR 2024) [10].
2) *Unsupervised*: SCPNet (ECCV 2024) [43], Alto (NIPS 2024) [13], SSHNet (CVPR 2025) [14] and SSHNet-D (CVPR 2025) [14], and CoLReg (Information Fusion 2025) [15].

4) Implementation Details: We adopt a single NVIDIA A6000 GPU to conduct all experiments, with a batch size of 16 for our networks. For $it = 0$, $\Psi_{\mathrm{ddpm}}^{\mathrm{M}}$ is trained for 200K iterations with learning rate of 2.5×10^{-4}, and $\mathcal{R}_s^D$ is trained for 120K iterations with a OneCycleLR scheduler up

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Algorithm 1 Training Procedure for CMDiff-Reg

Require: Unaligned Cross-modality Image Pairs $\{\mathcal{I}^{\mathrm{A}}, \mathcal{I}^{\mathrm{B}}\}$
Ensure: Registration Network $\mathcal{R}^{C}(\Phi_c^{(\mathrm{IT}-1)})$

for $it = 0$ to $\mathrm{IT}-1$ **do**
  if $it == 0$ **then**
    Train $\Psi^{\mathrm{M}}_{\mathrm{ddpm}}(\Phi^{\mathrm{M}(0)}_{\mathrm{d}})$ with $\mathbf{I}^{\mathrm{M,HN}}_{\mathrm{geo}}$ as $\mathbf{I}^{\mathrm{M}}_{\mathrm{geo}}$ by minimizing $\mathcal{L}_d^{\mathrm{d-M}}$, where $\mathrm{M} \in \{\mathrm{A}, \mathrm{B}\}$;
    Train $\mathcal{R}^{D}_s(\Phi^0_s)$ with self-supervised representations generated by **fixed** $\{\Psi^{\mathrm{A}}_{\mathrm{ddpm}}, \Psi^{\mathrm{B}}_{\mathrm{ddpm}}\}$ by minimizing $\mathcal{L}_s$;
    Load Φ^0_s to $\mathcal{R}^{C}$ and train $\mathcal{R}^{C}(\Phi^0_c)$ with pseudo-labels obtained by **fixed** $\mathcal{R}^{D}_s(\Phi^0_s)$ by minimizing $\mathcal{L}_c$;
  else
    Train $\Psi^{\mathrm{M}}_{\mathrm{geo}}(\Phi^{\mathrm{M}(it)}_{\mathrm{g}})$ by minimizing $\mathcal{L}_d^{g-\mathrm{M}}$, where $\mathrm{M} \in \{\mathrm{A}, \mathrm{B}\}$;
    Train $\Psi^{\mathrm{M}}_{\mathrm{ddpm}}(\Phi^{\mathrm{M}(it)}_{\mathrm{d}})$ with updated geometric features extracted by **fixed** $\Psi^{\mathrm{M}}_{\mathrm{geo}}$ by minimizing $\mathcal{L}_d^{\mathrm{d-M}}$, where $\mathrm{M} \in \{\mathrm{A}, \mathrm{B}\}$;
    Train $\mathcal{R}^{D}_s(\Phi^{it}_s)$ initialized from $\Phi^{(it-1)}_c$, with self-supervised representations generated by **fixed diffusion models and geometric extractors** $\{\Psi^{\mathrm{A}(it)}_{\mathrm{geo}}, \Psi^{\mathrm{B}(it)}_{\mathrm{geo}}\}$,$\{\Psi^{\mathrm{A}(it)}_{\mathrm{ddpm}}, \Psi^{\mathrm{B}(it)}_{\mathrm{ddpm}}\}$ by minimizing $\mathcal{L}_s$;
    Train $\mathcal{R}^{C}(\Phi^{it}_c)$ initialized from Φ^{it}_s, with pseudo-labels obtained by **fixed** $\mathcal{R}^{D}_s(\Phi^{it}_s)$ by minimizing $\mathcal{L}_c$;
  end if
end for

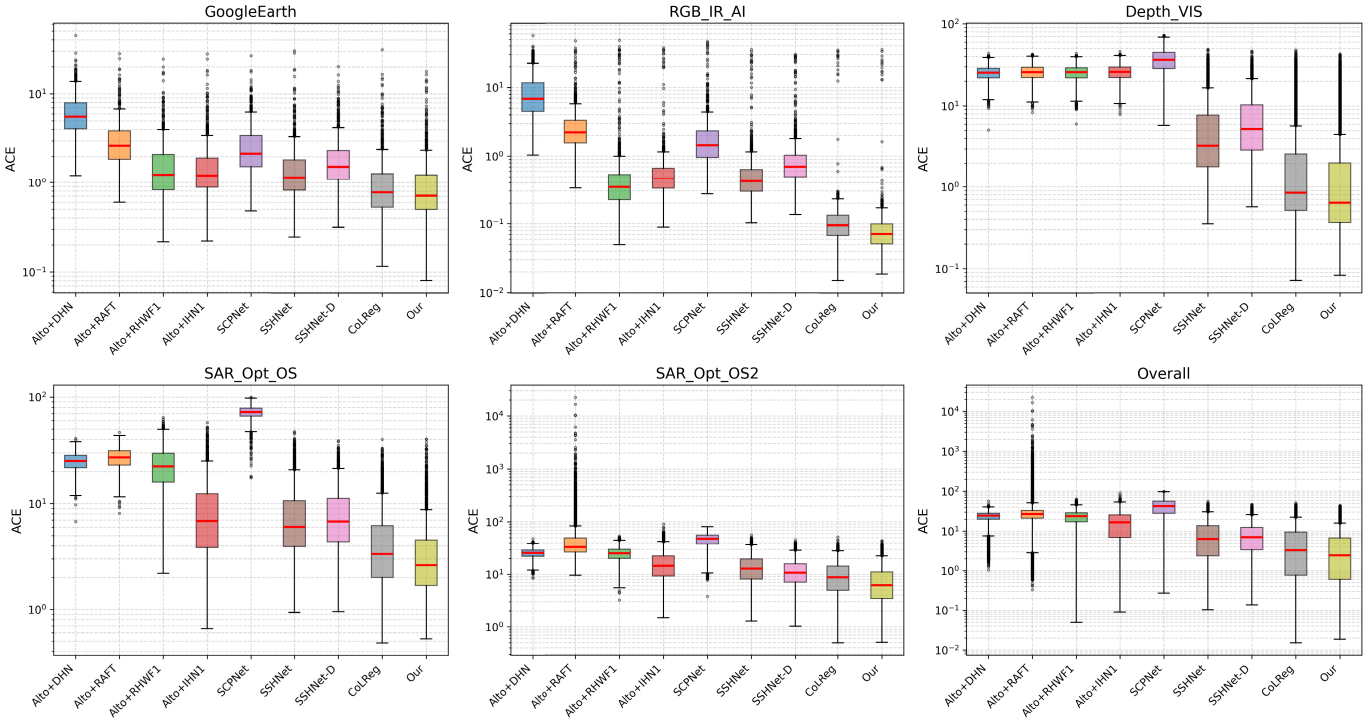


Fig. 6. Boxplots of ACE for different unsupervised image registration methods on five multimodal image registration datasets, including GoogleEarth, RGB_IR_AI, Depth_VIS, SAR_Opt_OS, and SAR_Opt_OS2. Each subfigure presents the distribution of ACE values across all test image pairs for a specific dataset, while the last subfigure summarizes the overall performance by combining all datasets together.

to 2.5×10^{-4}. For $it \geq 1$, $\Psi^{\mathrm{M}}_{\mathrm{geo}}$ and $\Psi^{\mathrm{M}}_{\mathrm{ddpm}}$, are trained for 30K and 60K iterations, respectively. For each of $\Psi^{\mathrm{M}}_{\mathrm{geo}}$, $\Psi^{\mathrm{M}}_{\mathrm{ddpm}}$, an optimizer is used with an initial learning rate of 2.5×10^{-4}. The learning rate schedule contains a short constant phase whose length decreases as *it* increases, followed by a linear decay to a smaller final value.

B. Comparison of Results

1) Quantitative Analysis: Fig. 6 presents a quantitative comparison between our method and other unsupervised deep learning approaches. On relatively easy datasets (GoogleEarth and RGB_IR_AI), CMDiff-Reg consistently outperforms most unsupervised methods. This advantage arises from its use of DDIR to construct challenging self-supervised cross-modal pairs, which encourage the network to extract discriminative, modality-invariant structural features. In contrast, prior methods such as SSHNet rely on monomodal self-supervised pairs, which are more sensitive to residual modality differences and may lose high-frequency geometric details during training. On more challenging datasets (Depth_VIS, SAR_Opt_OS, and SAR_Opt_OS2), CMDiff-Reg maintains its leading performance by leveraging geometry-enhanced conditional diffusion intermediate representations. As shown in Fig. 7, these representations act as an auxiliary modality that preserves some geometric structures while exhibiting significant modality

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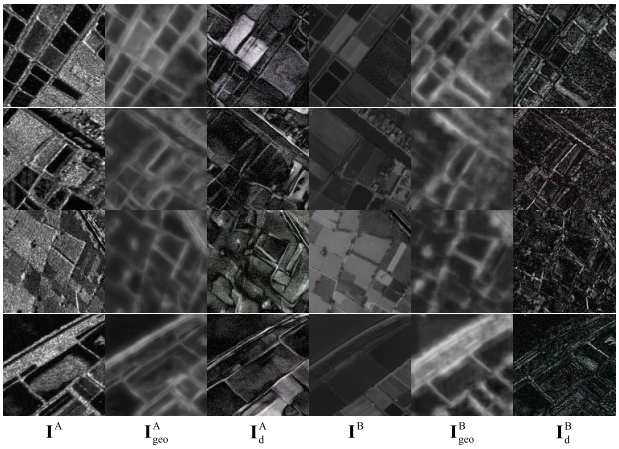


Fig. 7. Visualization of DDIRs $\mathbf{I}^{M}_{d}(M \in \{A, B\})$.

differences from the original images, providing stronger and more reliable self-supervision. This enables the registration network to handle large cross-modal variations more effectively than existing unsupervised methods, ensuring stable training and robust correspondence estimation in highly challenging scenarios.

In addition, as shown in Fig. 8, compared with supervised approaches, CMDiff-Reg approaches the upper bound established by methods such as MCNet. While a performance gap remains on the most challenging datasets, the combination of denoised-representation self-supervision, geometry enhancement, and stable training enables CMDiff-Reg to achieve near-supervised performance on easier datasets.

In previous comparisons, all methods—both supervised and unsupervised—were trained using pregenerated unaligned cross-modality image pairs for consistency. However, for supervised methods, when aligned cross-modality image pairs are available in the training set, unaligned pairs can be dynamically generated during training by applying random geometric transformations. To ensure a fairer comparison, in Table I, we compare our CMDiff-Reg with supervised methods (DHN*, IHN1*, IHN*, RHWF1*, RHWF*, and MCNet*) trained using this dynamic generation strategy. Even when the advantages of supervised methods are fully leveraged, our unsupervised CMDiff-Reg still outperforms the supervised methods DHN*, IHN1*, and RHWF1* on the easy dataset RGB_IR_AI, and achieves performance comparable to IHN*, RHWF*, and MCNet*. On the more challenging datasets Depth_VIS, SAR_Opt_OS, and SAR_Opt_OS2, our method still surpasses several supervised approaches on some metric evaluations.

2) Qualitative Analysis: On more challenging datasets (Depth_VIS, SAR_Opt_OS, and SAR_Opt_OS2), the source and target images exhibit substantial modality differences, not only in radiometric and geometric properties but also in object textures, structural details, and overall appearance. CMDiff-Reg leverages geometry-enhanced DDIRs to construct self-supervised cross-modal training pairs, which provide geometric information while introducing challenging modality variations. This enables the registration network to train stably and learn robust correspondences under large modality discrepancies. As illustrated in Fig. 9, on Depth_VIS, CMDiff-Reg consistently achieves superior alignment across diverse scenes and spatial locations. On SAR_Opt_OS and SAR_Opt_OS2, where the source and target images differ greatly in imaging mechanisms and geometric properties, CMDiff-Reg outperforms existing unsupervised deep learning approaches, demonstrating its robustness and generalization capability in highly challenging cross-modal registration tasks.

TABLE I

COMPARISON BETWEEN OUR UNSUPERVISED METHOD (CMDIFF-REG) AND SEVERAL SUPERVISED BASELINES, INCLUDING DHN*, RHWF1*, RHWF*, IHN1*, IHN*, AND MCNET*. THE SUPERVISED METHODS ARE TRAINED USING UNALIGNED CROSS-MODAL IMAGE PAIRS THAT ARE RANDOMLY GENERATED ON-THE-FLY DURING THE TRAINING PROCESS

RGB_IR_AI					
Method	AUC@3	AUC@5	AUC@10	AUC@20	MACE
DHN*	0.59	5.79	33.55	63.92	7.23
IHN1*	88.86	92.81	95.82	97.44	0.57
RHWF1*	92.90	95.22	96.99	98.07	0.43
IHN*	94.93	96.43	97.62	98.35	0.37
RHWF*	96.33	97.32	98.06	98.57	0.32
MCNet*	97.21	97.81	98.28	98.64	0.38
Our	96.09	97.18	97.99	98.52	0.34
Depth_VIS					
Method	AUC@3	AUC@5	AUC@10	AUC@20	MACE
DHN*	0.00	0.37	9.46	42.07	11.77
IHN1*	47.50	63.40	77.98	87.30	2.67
RHWF1*	56.91	69.44	81.06	88.82	2.36
IHN*	61.18	72.43	82.71	89.58	2.20
RHWF*	65.05	74.82	83.87	90.25	2.08
MCNet*	64.21	72.28	80.84	87.94	2.60
Our	60.71	68.95	77.88	85.38	3.22
SAR_Opt_OS					
Method	AUC@3	AUC@5	AUC@10	AUC@20	MACE
DHN*	0.35	3.52	22.54	54.37	9.28
IHN1*	16.51	38.10	64.59	80.79	3.95
RHWF1*	0.19	1.65	13.39	42.59	11.80
IHN*	19.94	41.32	66.14	81.51	3.84
RHWF*	21.10	43.63	68.09	82.07	3.83
MCNet*	26.21	48.82	71.84	84.97	3.08
Our	22.38	41.15	64.31	80.12	4.11
SAR_Opt_OS2					
Method	AUC@3	AUC@5	AUC@10	AUC@20	MACE
DHN*	0.01	0.23	5.57	34.03	13.68
IHN1*	2.27	12.03	39.67	65.22	7.24
RHWF1*	0.03	0.56	7.37	35.07	13.41
IHN*	4.75	17.45	44.17	66.86	7.19
RHWF*	0.06	0.59	8.74	37.43	12.88
MCNet*	7.69	22.56	50.56	72.92	5.57
Our	6.88	16.50	37.12	60.84	8.18

C. Time and Memory Cost

1) Training: Table II summarizes the training cost of CMDiff-Reg. The conditional diffusion model exhibits the highest GPU memory consumption among all components, while the remaining modules require relatively moderate GPU resources. Since all components are optimized sequentially rather than jointly, the overall GPU requirement is determined by the most memory-intensive stage rather than the cumulative memory consumption of all modules.

Fig. 10 illustrates the effect of different co-training iterations on the final test performance. Increasing the number of

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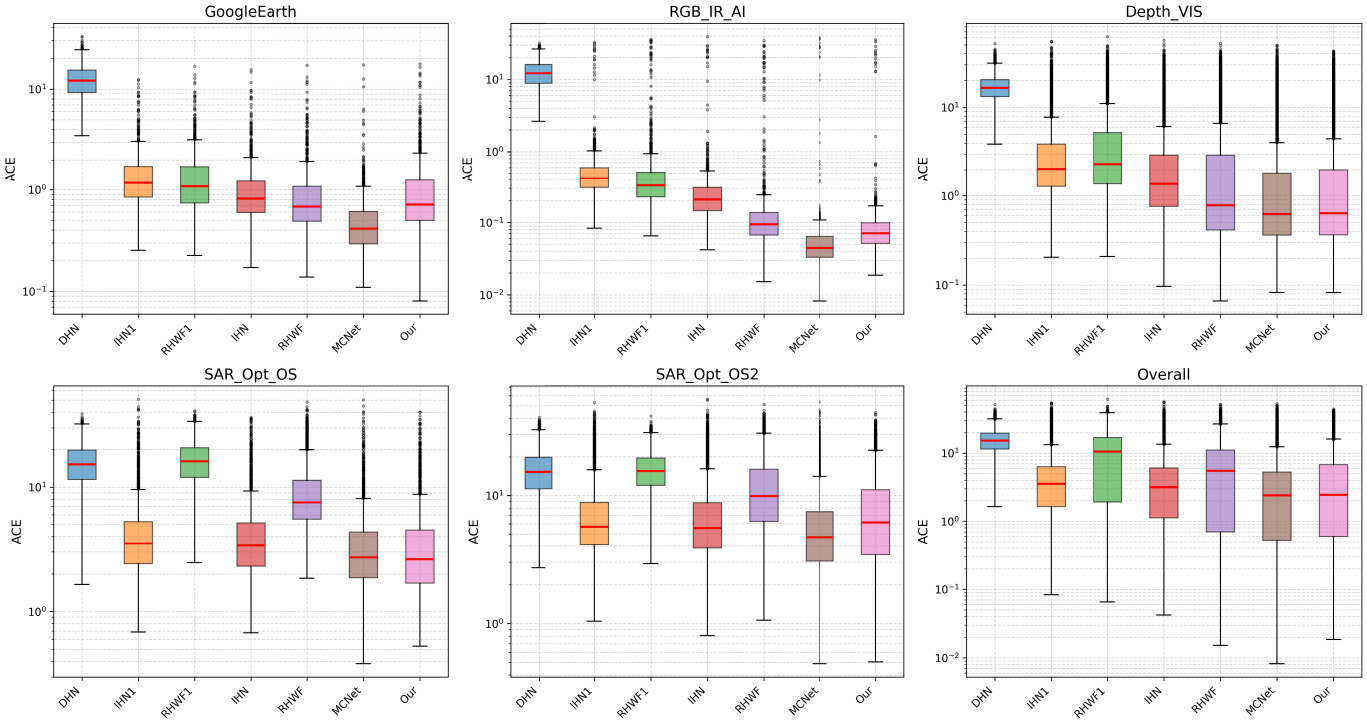


Fig. 8. Boxplots of ACE for different supervised image registration methods and our unsupervised method on five multimodal image registration datasets, including GoogleEarth, RGB_IR_AI, Depth_VIS, SAR_Opt_OS, and SAR_Opt_OS2.

TABLE II

TRAINING COST ANALYSIS OF CMDIFF-REG ON A SINGLE NVIDIA RTX A6000 GPU

Component	GPU Memory (GB)
Geometric condition extractor Ψ_{geo}	11
Conditional diffusion model Ψ_{ddpm}	15
Self-supervised registration network $\mathcal{R}_s^D$	9
Cross-modal registration network $\mathcal{R}^C$	5
Co-training Iterations	**Time (h)**
IT=1	57
IT=2	84
IT=3	111

TABLE III

INFERENCE TIME AND MEMORY USAGE OF DIFFERENT UNSUPERVISED REGISTRATION METHODS

Method	Time(ms)	GPU(GB)	CPU(GB)	Params(M)
SCPNet	12.51	0.98	3.39	1.60
AltO+DHN	7.14	0.84	3.52	21.30
AltO+RAFT	35.52	0.87	3.29	5.26
AltO+RHWF1	90.34	0.89	3.52	0.92
AltO+IHN1	22.88	0.78	3.51	1.29
SSHNet	50.07	0.99	4.33	8.25
SSHNet-D	27.47	0.79	4.30	0.74
CoLReg	42.49	1.01	3.60	1.20
Our	42.49	1.01	3.60	1.20

co-training iterations consistently improves registration accuracy across different datasets, while the performance gain gradually saturates. Therefore, the adopted setting provides a favorable tradeoff between training cost and registration performance.

2) Inference: As shown in Table III, CMDiff-Reg achieves superior registration performance compared with recent unsupervised methods while maintaining competitive inference efficiency. Since only the distilled registration network is employed during testing, the proposed framework introduces no additional diffusion-related computations during inference, demonstrating an effective balance between accuracy and computational cost.

D. Ablation Analysis

1) Diffusion Timestep Selection: We analyze the sensitivity of CMDiff-Reg to diffusion timesteps used for generating self-supervised representations. As shown in Fig. 11, early diffusion timesteps (e.g., 1–200) yield better performance, while late timesteps lead to noticeable degradation. This is because our self-supervised representations are constructed by comparing noise predictions before and after introducing cross-modal geometric conditions, rather than directly reconstructing images. Early timesteps correspond to lower noise levels, where the interaction between noised inputs and geometric conditions introduces stronger structural perturbations, making the task more challenging and encouraging more robust geometric correspondence learning. In contrast, late timesteps produce weaker perturbations and easier optimization, but are less effective for capturing complex cross-modal geometry, resulting in degraded registration performance.

2) Effectiveness of Unsupervised Multimodal Registration Framework: To evaluate the performance of CMDiff-Reg, we compared it with modified versions of

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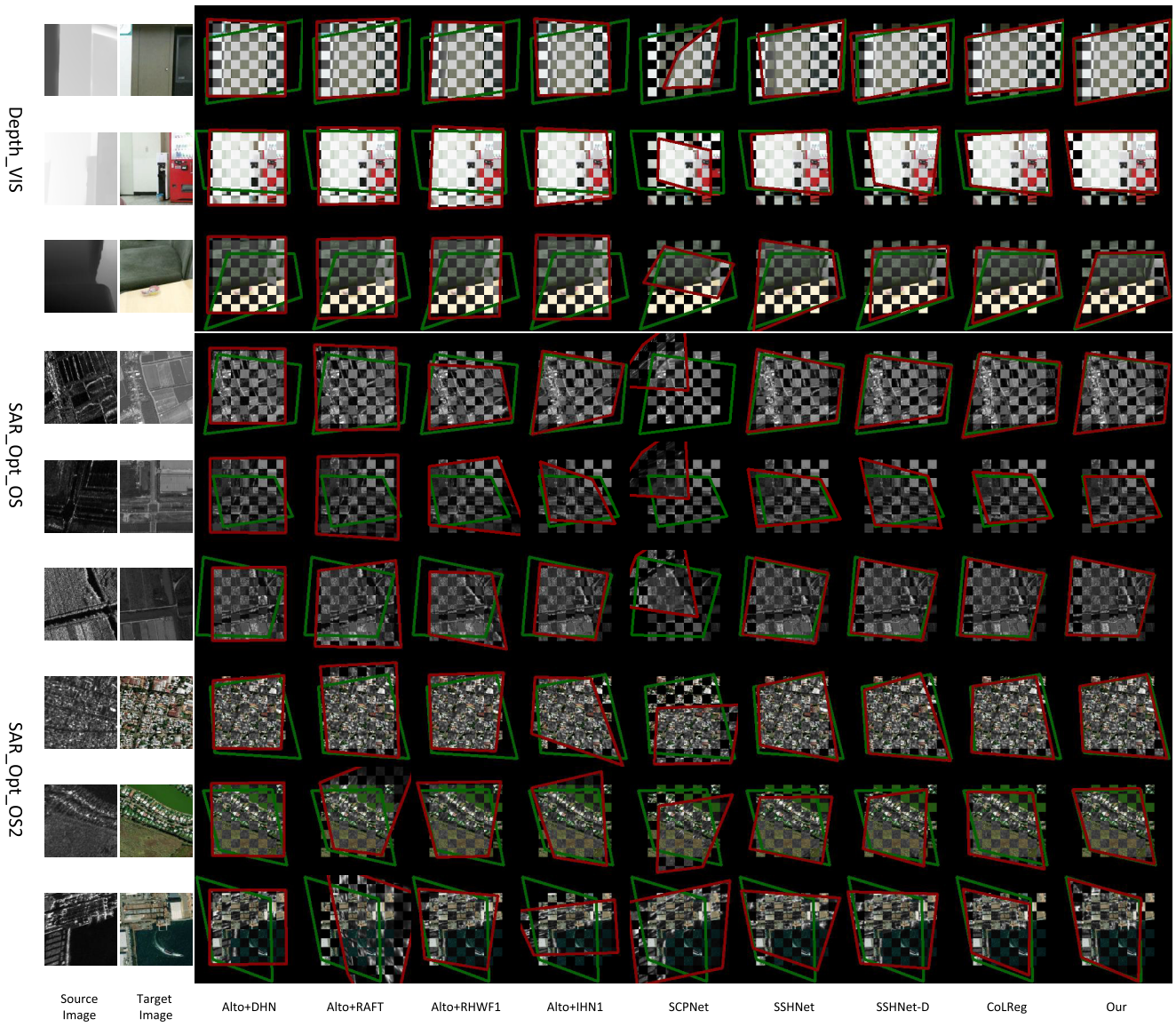


Fig. 9. Qualitative image registration results of our method CMDiff-Reg and other unsupervised methods: AltO+DHN, Alto+RAFT, AltO+RHWF1, AltO+IHN1, SCPNet, SSHNet, SSHNet-D, and CoLReg on the challenge dataset. Green polygons denote the ground-truth deformation from source images to target images. Red polygons denote the estimated deformation using different methods on the target images.

existing unsupervised registration frameworks, where the registration networks in SSHNet and SSHNet-D were replaced with the network used in our method, denoted as SSHNet*, and SSHNet-D*, as summarized in Table IV. The advantages of CMDiff-Reg are more pronounced on challenging datasets, including Depth_VIS, SAR_Opt_OS, and SAR_Opt_OS2. On Depth_VIS, CMDiff-Reg substantially improves all AUC metrics and reduces MACE compared to SSHNet* and SSHNet-D*. On SAR_Opt_OS and SAR_Opt_OS2, CMDiff-Reg achieves the best performance across all metrics, indicating strong robustness under large modality differences. These results demonstrate that the proposed framework effectively enhances registration accuracy, particularly in challenging multimodal scenarios.

3) Ablation Analysis of Loss Functions and Framework Modules: Table V(a) analyzes the effects of the three geometric losses used to optimize the geometric condition extractor. Removing any individual loss function results in performance degradation, indicating that each geometric constraint contributes to the overall effectiveness of the proposed framework. These results indicate that both the cross-modal geometric consistency constraint and the guidance effectiveness constraint play critical roles in learning reliable geometry-aware conditions. Furthermore, the explicit geometric structure alignment loss $\mathcal{L}_d^{g3\text{-M}}$ contributes positively. When all three losses are jointly employed, the model achieves the best performance, suggesting that the proposed losses provide complementary constraints for geometry-enhanced representation learning.

Table V(b) further evaluates the contribution of each major module in CMDiff-Reg. Without Ψ_{geo} and Ψ_{ddpm} yields poor performance. Introducing either the geometric condition extractor Ψ_{geo} or the conditional diffusion model Ψ_{ddpm} significantly improves registration accuracy. Combining these two components further improves performance, which validates the effectiveness of the proposed DDIRs in generating self-supervised training samples. Finally, incorporating the

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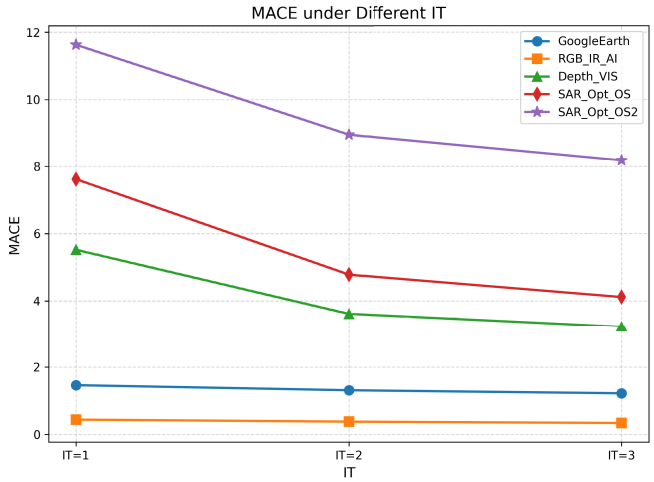


Fig. 10. Effect of the number of co-training iterations (IT) on the MACE across different datasets.

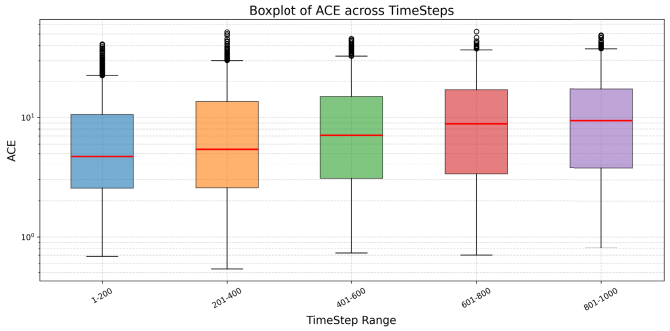


Fig. 11. Analysis of timesteps for generating self-supervised representations on SAR_Opt_OS with IT = 1.

knowledge distillation stage through $\mathcal{R}^C$ achieves the best overall performance. This result suggests that the distilled network can effectively transfer the geometric knowledge learned from diffusion-based self-supervision to real cross-modal image pairs. Overall, the ablation studies verify that each designed component contributes positively to the final performance and collectively demonstrate the effectiveness of the proposed framework.

4) Effectiveness of Proposed Denoised Self-Supervised Intermediate Representations: To verify the effectiveness of our proposed denoised self-supervised intermediate representations, we evaluate three different self-supervised representations, as illustrated in Fig. 12. The results clearly show that the self-supervised registration network trained with $\{\mathbf{I}^{A'}, \mathbf{I}_d^A\}$ and $\{\mathbf{I}^{B'}, \mathbf{I}_d^B\}$ achieves the highest registration accuracy, demonstrating that the denoised intermediate representations produced by our geometric-enhanced diffusion models serve as a more reliable auxiliary-modality self-supervision.

E. Zero-Shot Analysis

We design a comprehensive set of zero-shot cross-modality image registration experiments to evaluate the cross-domain generalization of various deep learning methods. As illustrated in Fig. 13, each column corresponds to a specific training dataset (i.e., GoogleEarth, RGB_IR_AI, Depth_VIS, SAR_Opt_OS, and SAR_Opt_OS2), while each row represents testing on unseen datasets from different modalities. This configuration allows us to assess how well each model adapts to previously unseen distributions without any fine-tuning.

Overall, our approach demonstrates robust generalization ability across all dataset pairs. When trained on visually rich datasets such as GoogleEarth or RGB_IR_AI, our method achieves a significantly higher proportion of high-precision registrations compared to other unsupervised approaches. Even

TABLE IV

ABLATION ANALYSIS OF DIFFERENT UNSUPERVISED IMAGE REGISTRATION FRAMEWORKS WITH THE REGISTRATION NETWORK USED IN OUR CMDIFF-REG

Method	GoogleEarth				
	AUC@3	AUC@5	AUC@10	AUC@20	MACE
SSHNet*	64.36	75.23	85.68	92.29	1.56
SSHNet-D*	54.36	67.37	81.38	90.44	1.91
CoLReg	64.57	76.68	87.47	93.55	1.30
Our	**66.34**	**77.94**	**88.14**	**93.88**	**1.22**
Method	RGB_IR_AI				
	AUC@3	AUC@5	AUC@10	AUC@20	MACE
SSHNet*	95.21	96.35	97.34	97.95	0.50
SSHNet-D*	82.26	85.70	89.98	93.83	1.32
CoLReg	95.25	96.63	97.67	98.28	0.42
Our	**96.09**	**97.18**	**97.99**	**98.52**	**0.34**
Method	Depth_VIS				
	AUC@3	AUC@5	AUC@10	AUC@20	MACE
SSHNet*	28.31	42.93	60.35	74.89	5.35
SSHNet-D*	30.22	43.00	59.22	74.34	5.38
CoLReg	55.06	64.70	74.71	83.08	3.80
Our	**60.71**	**68.95**	**77.88**	**85.38**	**3.22**
Method	SAR_Opt_OS				
	AUC@3	AUC@5	AUC@10	AUC@20	MACE
SSHNet*	0.69	4.75	21.32	47.16	11.48
SSHNet-D*	0.65	4.53	23.26	52.84	9.73
CoLReg	16.31	32.84	56.09	74.73	5.20
Our	**22.38**	**41.15**	**64.31**	**80.12**	**4.11**
Method	SAR_Opt_OS2				
	AUC@3	AUC@5	AUC@10	AUC@20	MACE
SSHNet*	1.34	4.01	12.31	30.32	16.35
SSHNet-D*	1.20	4.43	17.69	44.02	11.84
CoLReg	2.78	8.94	25.55	50.98	10.30
Our	**6.88**	**16.50**	**37.12**	**60.84**	**8.18**

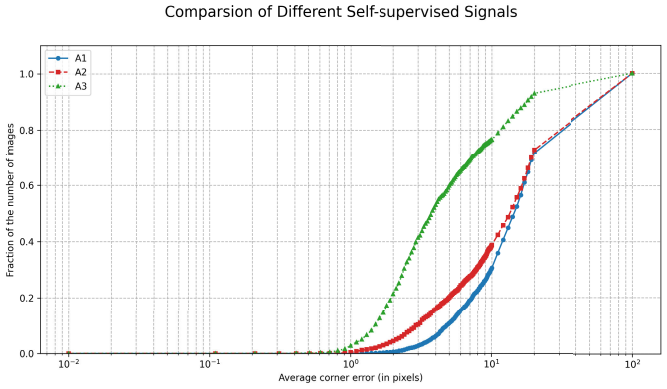


Fig. 12. Ablation study on different self-supervised signals. (A1) Using $\{\mathbf{I}^{A'}, \mathbf{I}^{A}\}$ and $\{\mathbf{I}^{B'}, \mathbf{I}^{B}\}$ to train self-supervised registration network. (A2) Using $\{\mathbf{I}^{A'}, \mathbf{I}^{A,HN}\}$ and $\{\mathbf{I}^{B'}, \mathbf{I}^{B,HN}\}$ to train self-supervised registration network. (A3) Using $\{\mathbf{I}^{A'}, \mathbf{I}_d^{A}\}$ and $\{\mathbf{I}^{B'}, \mathbf{I}_d^{B}\}$ to train self-supervised registration network.

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TABLE V

ABLATION STUDIES ON SAR_OPT_OS. THE FIRST PART EVALUATES THE GEOMETRIC LOSS FUNCTIONS FOR THE GEOMETRIC CONDITION EXTRACTOR, WHILE THE SECOND PART STUDIES THE CONTRIBUTION OF MODULES IN OUR UNSUPERVISED MULTIMODAL IMAGE REGISTRATION FRAMEWORK (A) GEOMETRIC LOSS FUNCTIONS AND (B) MODULES IN THE FRAMEWORK

(a) Geometric Loss Functions							
$\mathcal{L}_d^{g1\text{-M}}$	$\mathcal{L}_d^{g2\text{-M}}$	$\mathcal{L}_d^{g3\text{-M}}$	AUC@3	AUC@5	AUC@10	AUC@20	MACE
×	✓	✓	19.18	37.52	61.22	78.31	4.45
✓	×	✓	19.04	37.81	61.58	78.42	4.44
✓	✓	×	21.68	39.85	62.32	78.69	4.36
✓	✓	✓	**22.38**	**41.15**	**64.31**	**80.12**	**4.11**
(b) Modules in the Framework							
Ψ_{geo}	Ψ_{ddpm}	$\mathcal{R}^C$	AUC@3	AUC@5	AUC@10	AUC@20	MACE
×	×	✓	3.07	7.99	19.11	38.95	13.61
✓	×	✓	13.46	27.92	48.99	68.28	6.70
×	✓	✓	13.79	28.17	49.18	68.74	6.48
✓	✓	×	20.61	38.03	60.12	76.29	4.97
✓	✓	✓	**22.38**	**41.15**	**64.31**	**80.12**	**4.11**

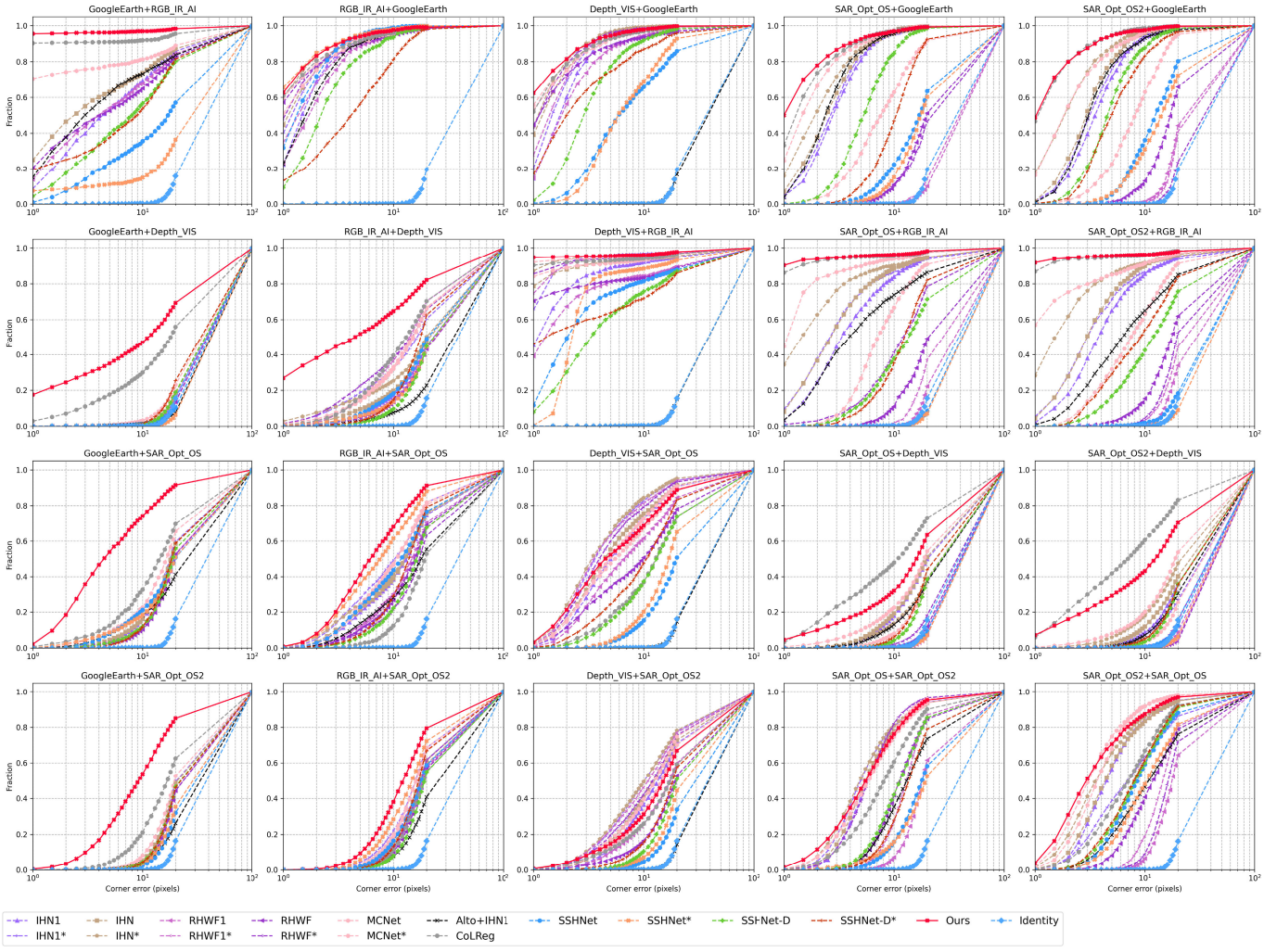


Fig. 13. Comparison of zero-shot cross-modal image registration performance across different training–testing dataset combinations. Each column corresponds to a specific training domain and each row represents testing on unseen modalities. Our method consistently achieves good registration accuracy and demonstrates superior cross-domain generalization.

when encountering the geometrically and radiometrically challenging Depth_VIS, SAR_Opt_OS, and SAR_Opt_OS2 datasets, our model consistently maintains strong registration accuracy in some cases. These zero-shot evaluation results indicate that CMDiff-Reg maintains strong generalization capability when applied to unseen datasets and modalities, demonstrating its robustness and adaptability in challenging cross-domain registration tasks.

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V. Conclusion

In this work, we presented CMDiff-Reg, an unsupervised framework for cross-modality remote sensing image registration that leverages DDIR as an auxiliary modality for self-supervision. By constructing challenging cross-modality training pairs, CMDiff-Reg enables the registration network to learn robust geometric representations and improves its ability to handle large modality discrepancies commonly encountered in remote sensing scenarios. To enhance the geometric reliability of the generated representations, we further proposed a geometry-enhanced conditional diffusion model that preserves structural information while introducing significant appearance differences. Compared with existing translation-based self-supervised approaches, the proposed diffusion-based strategy provides more stable supervision and reduces the effect caused by imperfect modality translation. Extensive experiments on multiple cross-modality datasets demonstrate that CMDiff-Reg consistently outperforms state-of-the-art unsupervised registration methods and achieves performance competitive with advanced supervised approaches. In particular, the proposed method shows strong robustness on challenging remote sensing scenarios with large modality gaps. Overall, these results highlight the effectiveness of DDIR as a reliable self-supervision mechanism for cross-modality remote sensing registration. We believe that the proposed framework provides a promising direction for robust cross-modality alignment and may benefit a wide range of remote sensing applications, such as multimodal data fusion, change detection, and geo-localization. We note that our current evaluation focuses on homography-based global transformations under controlled settings. Extending the proposed framework to more complex real-world scenarios involving nonrigid deformations and occlusions is an important direction for future work.

VI. Future Work

Despite the promising performance of the proposed method, several challenges remain for unsupervised cross-modality remote sensing image registration. First, real-world deployment still faces difficulties due to factors, such as sensor noise, resolution inconsistencies, temporal variations, and complex imaging conditions, which may affect the robustness and generalization ability of the model. Second, extending the framework to large-scale cross-sensor registration scenarios poses additional challenges in terms of computational efficiency, memory consumption, and scalability when handling high-resolution and large-volume remote sensing data. Third, integrating the proposed framework with emerging data-driven paradigms, such as foundation models and large-scale pretraining strategies, is a promising direction to further enhance cross-modal representation learning and improve generalization across diverse sensing modalities. Exploring these directions will be an important focus of our future work.

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